

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
5	(a)		- Placing of laser slits, screen or travelling microscope or transverse ruler in words or diagram even if no suggested distances. - Distance between slits and screen measured: stated with metre ruler or tape measure . - Fringe separation/spacing measured with ruler [or travelling microscope] with details included e.g. between centres of fringes or across several fringes or repeat readings taken or use of magnifying glass or large slit-screen distance - Slit-screen distance, slit separation, fringe separation ‘put into formula’ stated: $\lambda = \frac{\text{slit separation} \times \text{fringe separation}}{\text{slit-screen distance}}$ symbols, if used, must be identified with measured quantities. No penalty for swapping <i>a</i> and <i>y</i>. 5-6 marks Comprehensive description of how to set up the apparatus, the measurements to be taken (including details of instruments / techniques) and how to determine the wavelength with the equation clearly stated. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks Description of how to set up the apparatus, the measurements to be taken either both measurements specified or one measurement / technique and how to determine the wavelength with the equation stated. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i>	6			6		6

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				AO1	AO2	AO3	Total		
			1-2 marks Some correct attempt at either how to set up the apparatus, which measurements are taken and λ or how the wavelength is determined. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)	(i)	Right angled triangle drawn and labelled with 0.6[00] on the correct side (1) adj side $\approx$ hypot or $\sin \theta_1 \approx \tan \theta_1$ (accept calculations) or equivalent (1)		2		2	2	
		(ii)	Correct insertion of values of d , $\sin \theta_1$ and $n = 1$ in equation or correct use of Youngs slits equation (1) $\lambda = 510 \text{ nm}$ (1)		2		2	2	
		(iii)	39 orders (1) ecf accept 39.2 orders Slit separation is very large [compared with λ] (1) Accept λ <u>much</u> less than slit separation		1	1	2	1	
			Question 5 total	6	5	1	12	5	6